CSE558 · DATA SCIENCE — LECTURE 1

Welcome to Data Science

Housekeeping first — then a 170-year-old mystery that explains why we're all here.

TODAY'S ARC

Logistics → Origin story → The process → Your project

Supratim · *Monsoon-2026*

Claude & Gemini helped with slides.

HOW THIS COURSE RUNS

Course essentials

Schedule	Monday & Wednesday at 3:00–4:30 PM, in B003
Format	In-person; occasional hybrid sessions announced in advance
Office hours	Thursdays, second half of the day (Online / B-512)
Course page	bit.ly/Dsc2026 · class code: jsbnyord
Announcements	All posted in Google Classroom — check it regularly



SUBMISSIONS & SOLUTIONS

All evaluation components are submitted and graded in Google Classroom. Do not post solutions in Google Classroom or any public forum.

HOW YOUR GRADE IS CALCULATED

Evaluation & grading

Evaluation Type	Contribution
Assignment	**10%**
Quiz	**20%**
Mid-Sem	**20%**
End-Sem	**25%**
Project	**25%**

Grading policy: Relative

Relative grading, no fixed cutoff.
No fixed percentage that guarantees an A.

KEY POLICIES

Quizzes

Best n−1 counted. No makeup quiz under any circumstance.

Semester Exams

Makeup only for midsem & Endsem with approved medical leave.

Attendance

Not mandatory, however the surprise quizzes depend on it.

Project timeline

Groups finalized — within 2 weeks. Proposal — early September.
Progress — after mid-sem. Conclusion — before end-sem.



COLLABORATION, INTEGRITY, SUBMISSION

Ground rules



COLLABORATION

Discuss questions with others but acknowledge.
Group homework: discuss, but write your own version.
Group projects: TAs available for advice — maximum group size is 6.



SUBMISSION MECHANICS

Submit at least 4 hours before the deadline — GC issues in the last hour won't be considered.
Accepted format: PDF only (MS Word, Google Docs, or LaTeX).



ZERO TOLERANCE — THIS IS WHAT COSTS YOU THE MOST

PLAGIARISM

Assignment / Project — 1st instance: zero + one letter grade down. 2nd instance: F.

Quiz / Exam — 1st instance: F grade. No warning, no second chance.

LATE SUBMISSION

Day 1	**−10%**
Day 2	**−30%**
Day 3	**−60%**
Day 4	**−100% (no credit)**

Prerequisites

MANDATORY BACKGROUND

Introduction to Programming

Algorithm Design & Analysis

You must be available with designing Algorithms and implement them on python

DESIRABLE

Basic statistics

Mean, median, mode; variance, standard deviation

Basic probability

Probability space; dependent/independent events; random variables

Basic linear algebra

Subspace, null space, rank

Comfortable with mathematical notation will help.

FOR DEEPER READING

Textbooks & references

Probability & Computing	*Mitzenmacher & Upfal*
Foundations of Data Science	*Blum, Hopcroft & Kannan*
Mining of Massive Datasets	*Leskovec, Rajaraman & Ullman*
Data Mining	*Han, Kamber & Pei*
Lecture Notes on Randomized Linear Algebra	*Mahoney*

Sometimes, research papers will be referenced as supplementary reading.

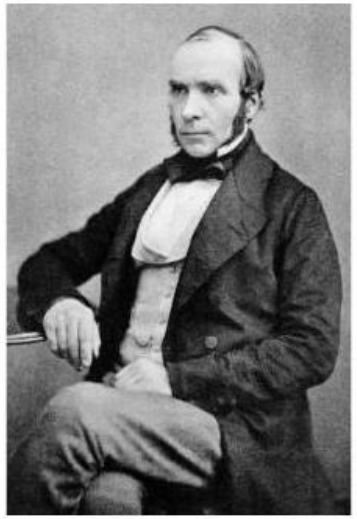
Slides and Reading materials are enough for the course.

A 170-YEAR-OLD DATA SCIENCE PROJECT

The doctor who didn't believe the air was poison



Miasma theory: disease from "bad air"



John Snow, 1813–1858

1817–1824	Asia, 350,000 died
1829–1837	Europe, 300,000 died
1846–1860	World, 2,000,000 died

Snow, a London physician, was not convinced. He suspected something in the water — not the air.

This is widely considered the first true data science project.

STATE OF EVENTS · 1

Investigate & collect data

Snow began his career in Newcastle, then built a well-connected London practice by 1850. He'd studied a smaller outbreak back in 1849 and already suspected water.

QUICK PROMPT *If you were the investigating physician, what's the very first question you'd ask a patient?*

ASKED QUESTIONS

Two patients, in the same hospital bed, died one after another.

First hypothesis: airborne — contagious from bed linens.

A new hypothesis takes shape

Cholera is not due to poison in the air.

May be contagious.

May be a water/food-borne disease.

SAMPLED A SET

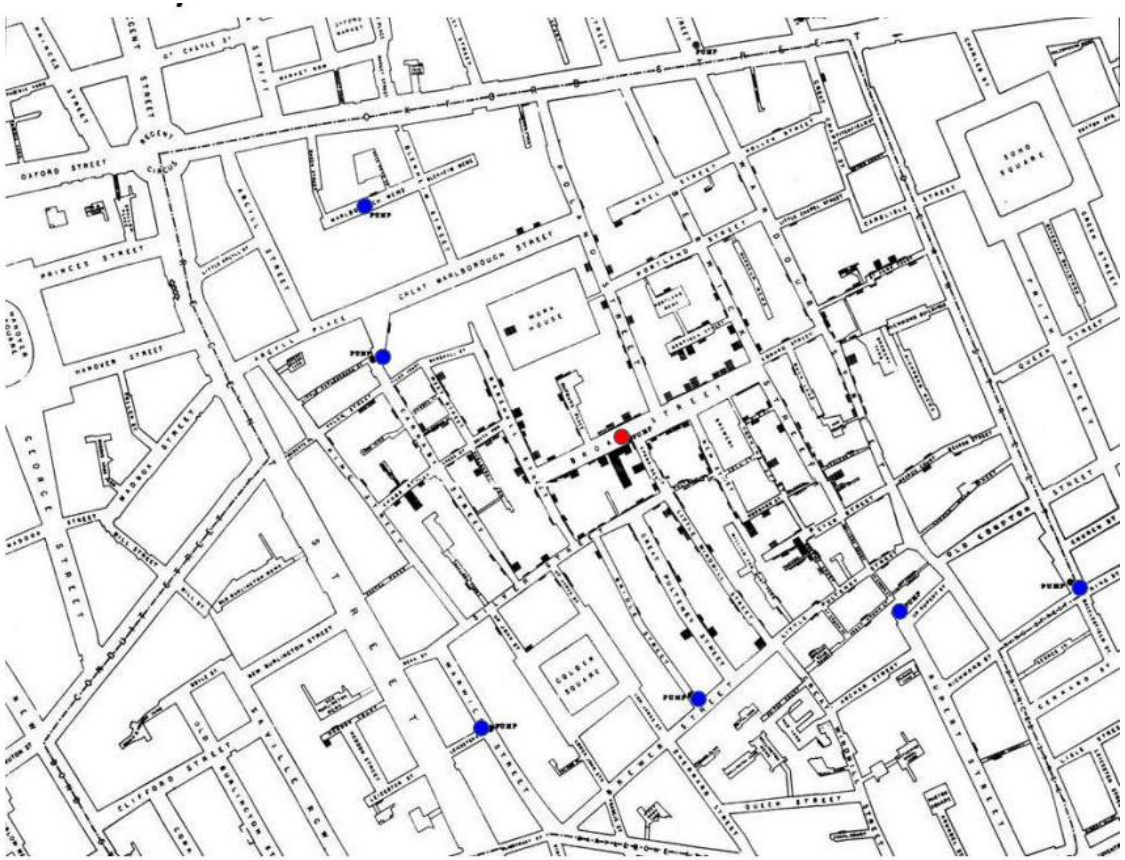
Yet, authorities did not believe him.

London's basic sewage system was failing to handle a rapidly rising population — raw sewage flowed through the streets and into the water supply.



Sewage seeping from cesspits toward nearby wells

Analyze data: the map that changed public health



Snow's map: each mark is a death. Red = the Broadway pump.

THE OUTBREAK, BY THE NUMBERS

Began 31 August 1854. In just 10 days, roughly 500 people in the immediate area died.

500 deaths a day at the outbreak's peak.

Snow plotted every recorded death by address on a map of Soho. A pattern immediately emerged around a pump on Broad Street.

Handle the outliers



Private pump — spared



Brewery workers — spared

An outlier isn't automatically noise. Sometimes it's the clue that confirms your theory instead of breaking it.

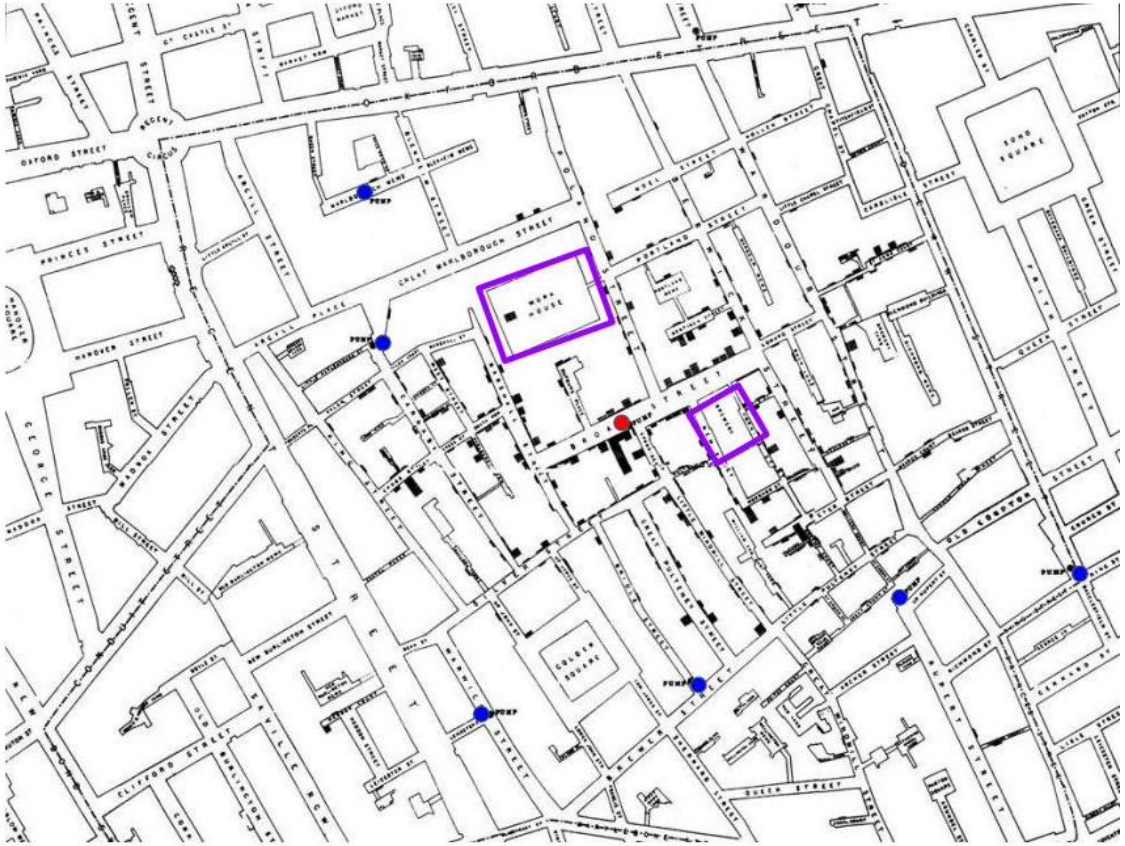
THE HUGGINS BREWERY, IN SNOW'S OWN WORDS

"There is a Brewery in Broad Street... on perceiving that no brewer's men were registered as having died of cholera, I called on Mr. Huggins, the proprietor. He informed me that there were above seventy workmen employed... none of them had suffered from cholera."

— John Snow, On the Mode of Communication of Cholera, 1855

The workmen drank the malt liquor they made, or water from the brewery's own well — never from the Broadway pump. What looked like a gap in the pattern was actually confirmation of it.

The trained model



Purple: the workhouse (mostly spared) and the brewery

CONCLUSION

People are dying by consuming water from the Broadway street pump.

The workhouse on Poland Street had its own well — its residents were also largely spared, another confirming outlier, not a contradiction.

Validate: stop using the well

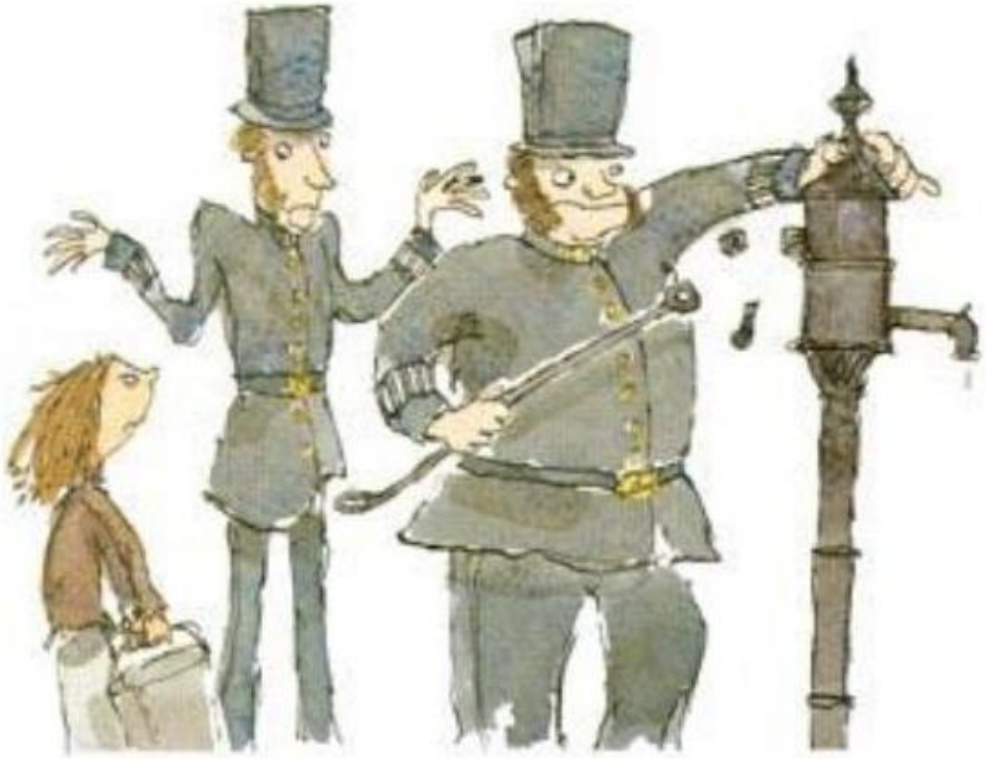


Illustration: the pump handle is removed

8 SEPTEMBER 1854

Snow convinced the local parish council to remove the Broad Street pump handle — a single, physically irreversible test of his theory.

"Removal of the pump handle" has since become an international symbol of public-health intervention — cited in every epidemiology textbook that follows.

Validate: the results come in

0 deaths per day, within a week.

A dramatic, immediate result.

THE MIASMA CAMP'S OBJECTION

"What about the false negatives — people who were far from the pump, yet still got sick?"

Further investigation revealed that for those people used to travel via broadway street and had drank water from the street pump.



The pump water, later confirmed contaminated



WHAT SNOW'S MAP TEACHES US

The data science process

Five steps, one loop — every technique in this course exists to make one of these five stronger, faster, or more rigorous.*

1. **Understand the Problem** →
2. **Data Collection** →
3. **Data Preparation** →
4. **Plan, Train & Validate Model** →
5. **Data Insight**

Snow ran all five steps in 1854, by hand. This course teaches you to run the same loop — faster, at scale, and with a rigorous measure of how confident to be.



THE SAME FIVE STEPS, IN 2026

A modern echo: predicting stroke risk

Collect	Patient history — blood pressure, cholesterol, diabetes, height, weight, age.
Prepare	Handle outliers, reduce dimensions (e.g. combine height & weight into BMI).
Plan & train	Build a decision tree model or some other classifier.
Validate	Cross-validation.
Deploy	Once validated, roll out to production.

From a hand-drawn map to a trained model — same loop, new tools.



BEFORE WE GO FURTHER — YOUR TOOLKIT

What you'll actually use this semester

An IDE

Jupyter, VS Code, or Colab — pick one and get comfortable. This is where you'll write, run, and debug every line of code this semester.

Python basics

Variables, loops, conditionals, functions, reading files. If you can write a for-loop and read a CSV, you're ready to start.

pandas, sklearn

DataFrames, cleaning, groupby/merge, handling missing data, modeling data. These will be the most-used libraries.

Visualization

matplotlib — charts, histograms, scatter plots.



YOUR PROJECT STARTS TODAY — STEPS 1 & 2

Understand the problem, then collect data

1. Understand the problem

Frame the problem. What are the key stakes? What are the objectives — business needs or research goals? What are the risks?

2. Collect data

Plan how you'll collect or generate data — sensors, surveys, crowdsourcing, scraping, existing sources. Get a high-level understanding before committing. Collection is expensive: start with a small sample and a low-stakes hypothesis.

STEPS 3, 4 & 5

Prepare, train, and land on insight

3

Prepare data

Convert text/graphs/sound to numbers, remove outliers, handle missing or inconsistent values, handle scale.

4

Plan, train & validate

Visualize initial patterns, understand correlations, draw initial conclusions, choose and tune a model, then validate it.

5

Data insight

Analyze what the model predicts, draw conclusions, show tables or plots that support your claims.

BEFORE WE MOVE ON — THE CLOCK STARTS NOW

Form your group. Start exploring.

1 THIS WEEK

Start talking

Look for classmates whose skills or interests complement yours — someone who codes, someone who knows the domain, someone who asks good questions.

2 WITHIN 2 WEEKS

Finalize your group

Lock in your team. Maximum group size: 6 people.

3 EARLY SEPTEMBER

Submit your proposal

A clear problem statement, framed using the process from today's lecture. This is graded.

WHAT MAKES A GOOD GROUP?

A mix of skills, not a friend group. · A problem you're genuinely curious about. · A size you can actually coordinate — 6 is a ceiling, not a target.

Snow didn't wait for permission to start investigating. The best projects start with curiosity, not certainty — start talking to people today.



13 WEEKS, ONE THROUGHLINE

The journey ahead

Week 1

Foundations

Data & descriptive statistics — where we are today

Wks 2–3

Inference

Probability & hypothesis testing

Wks 4–7

Concentration & streaming

Tail bounds, hashing, sketches

Wks 8–11

High-dimensional data

JL, Dimensionality Reduction, low-rank & matrix methods

Wks 12–13

Applications

Regression, clustering, your final project



THINK · PAIR · SHARE

You are already a dataset

With the person next to you, list 3 data points about yourself that a company could plausibly record about you today.

Behavioral

e.g. how much time do you spend on social media

Biometric

e.g. your heart rate from a fitness tracker

Preference

e.g. your last 5 search queries

Transactional

e.g. your last purchase amount

WHY WE'RE DOING THIS

Every one of those data points has a type — that's exactly where Lecture 2 begins.

A SNEAK PREVIEW

Your "Behavioral" example is probably continuous. "Preference" is probably categorical. By next lecture, you'll be able to say exactly why.

WRAP-UP

Before you go

ONE-MINUTE PAPER

1. Still any open questions in your mind about course logistics

2. One question you'd ask John Snow if he were here

Before your next class: talk to at least one classmate about your project?

NEXT LECTURE

The Vocabulary of Data

Objects, attributes, and the categories that describe them — plus your first statistical tool: central tendency (mean, median, mode).

TAKEAWAY TO CARRY FORWARD

"Every dataset has at least one story. Your job is to understand the story by following the same five steps Snow used — just with better tools, and tell the story to everyone."